ROS 2 Lab Report Delivery Tips

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***Abstract* — We outline a few rules to adhere to in order to properly prepare your labs. The main programming languages you are going to use to implement some control algorithms in ROS 2 are: Python, C++ and shell scripting using GNU Bash. It is preferable to write your lab reports in Typst, given the provided files.**

***Index terms* — ROS 2, Python, C++, GNU-Bash, Typst**

I. Programming Languages

The programming languages we will use in this module are Python (Figure 1), C++ (Figure 2), and shell scripting with GNU Bash (Figure 3). They are chosen for their expressive syntax, performance where needed, and extensive ecosystems.

Python Rapid development and high-level logic.
C++ Performance and control over system resources.
GNU Bash Scripting and automation of tasks.

Figure 1: Python logo

Figure 2: C++ logo

Figure 3: GNU Bash logo

Insert your code in each lab report. The detailed explanation of the objects, along the packages you have used is a must. After each code snippet, add a screenshot that showcases your running work when applying the test provided in each exercise.

II. Typst

Consider using Typst to write your lab reports. The provided templates allow you to focus on the content and seamlessly create a professional-looking report.
Typst supports Markdown-like syntax, which provides a range of formatting options [1]. Here are some points to help you write your report:

1. Formatting text:
   - Bold: surround text with asterisks '(*)'
   - Emphasis: surround text with underscores '(_)'
   - Headings: start a line with one or more equal signs '(=)'
2. Lists:
   - Unordered: start with a hyphen '(-)'
   - Ordered: start with a plus '(+)'
3. Inserting Objects:
   - Use this syntax if you need to draw a table:

```
#figure(
  table(
    columns: 4,
    [Row 1], [a], [b], [c],
    [Row 2], [1], [2], [3],
  ),
  caption: [Results],
) <tab:LABEL>

@tab:LABEL displays some results.
```

   - Use this syntax if you need to insert an image:

```
#figure(
  image("IMAGE_NAME.EXT", width: 100%),
  caption: [IMAGE_CAPTION],
) <fig:LABEL>

@fig:LABEL shows an image.
```

4. Code snippets:
   - Inline: wrap code with backticks (`)
   - Block: use triple backticks with a language label (e.g., 'python')

```python
import rclpy
from rclpy.node import Node
```

In each document, you have to insert well annotated screenshots of your code after being executed.

You can leverage those features using the app's intuitive interface and the provided template at the url https://typst.app/universe/package/ailab-isetbz, as shown in Figure 4. No installation is required; however, you may need to sign up to use the online editor. Keep an eye on your project size. Do not exceed 200MB. A fully fledged documentation on the usage of Typst is available at https://typst.app/docs/.

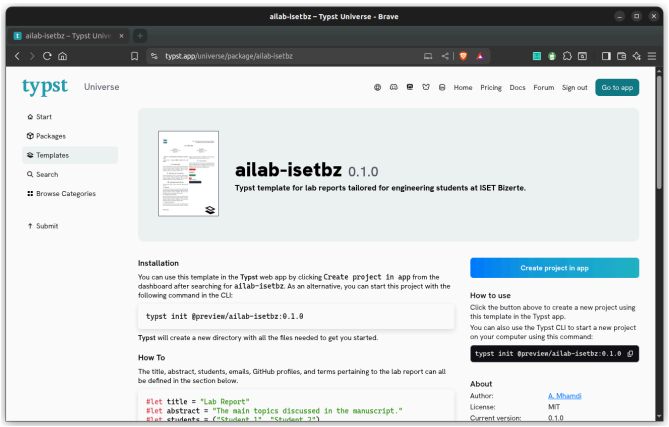


Figure 4: Typst app

III. GitHub

Share your code on GitHub. It's a fantastic way to foster a supportive coding community while gaining exposure to different coding styles and techniques [2], [3], [4].

IV. Links Bundle

You may find the following links useful:

- GitHub Repository (*Figure 5*) https://github.com/a-mhamdi/ros2

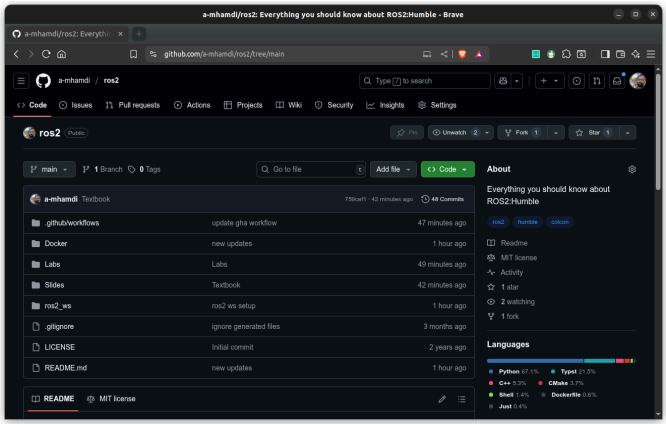


Figure 5: GitHub repository

- Docker Image (*Figure 6*) hub.docker.com/repository/docker/abmhamdi/ros2

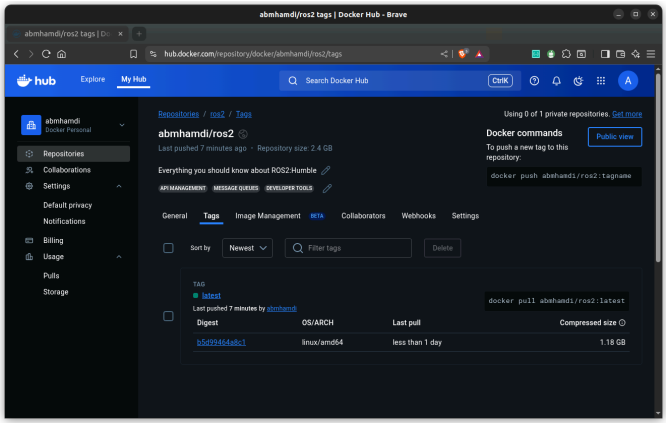


Figure 6: Docker image

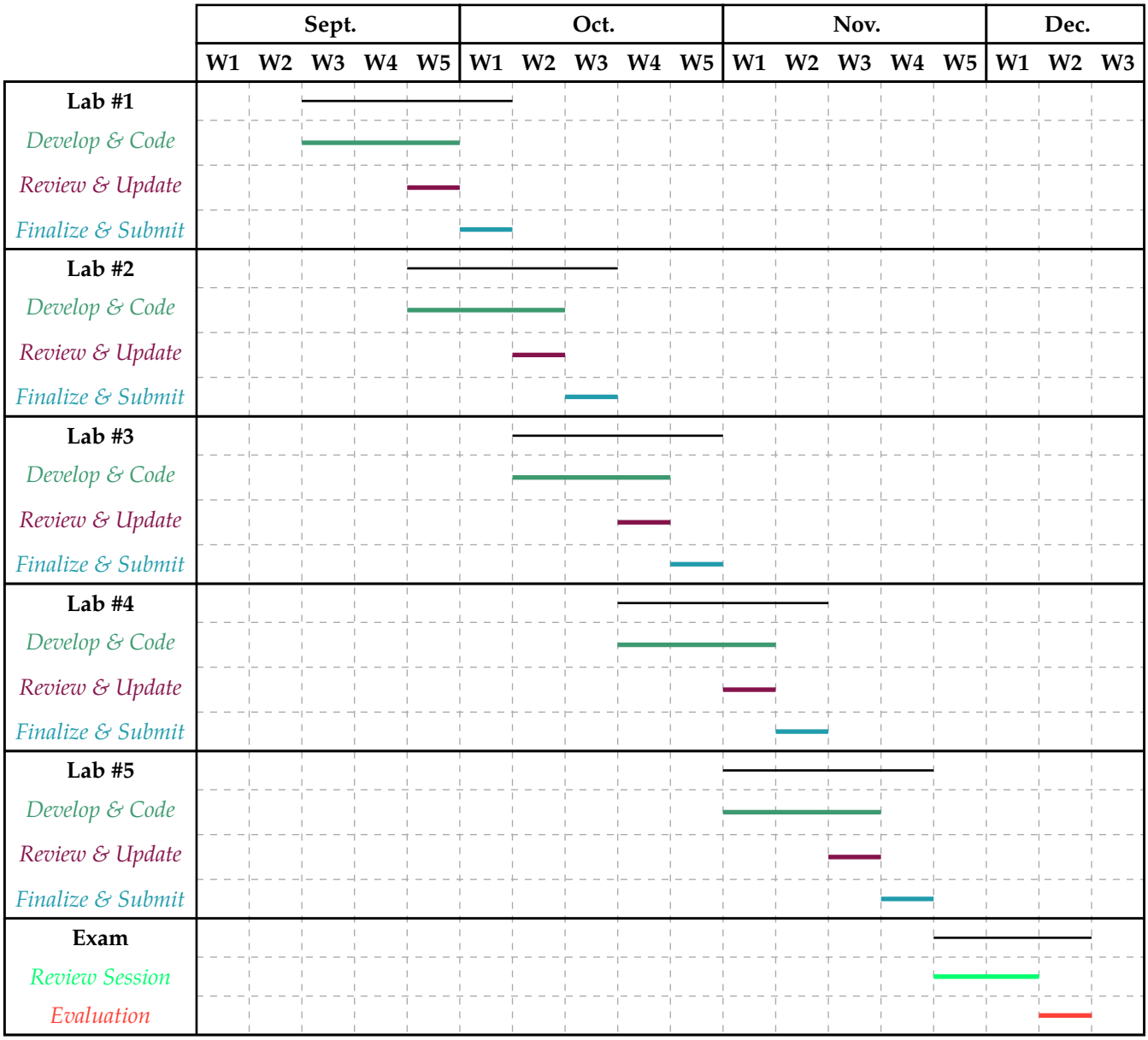
V. Submission Checklist

Before submitting your lab report, verify:

- Code builds and runs (include exact commands)
- Console output and screenshots accompany each task
- Public repository link is provided
- Document uses the Typst template with proper captions for figures/tables

VI. Timeline

The following timeline is proposed to help you organize your work. It is not mandatory to follow it, but it is highly recommended to do so. The labs are designed to be completed in the order they are presented.



References

[1] T. Mailund, *Introducing Markdown and Pandoc*. Berkeley, CA: Apress L. P., 2019.

[2] S. Guthals, *GitHub*, 2nd edition. in For dummies. Hoboken, NJ: John Wiley & Sons, Inc., 2023.

[3] P. Bell and B. Beer, *Introducing GitHub*, 1. ed. Beijing: O'Reilly, 2015.

[4] M. Tsitoara, *Beginning Git and GitHub*. Berkeley, CA: Apress, 2020.